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Module 3

IoT Protocols

1 IEEE 802.15.4

- IEEE 802.15.4 is a wireless access technology for **low-cost** and **low-data-rate devices** that are powered or run on batteries.
- This access technology enables **easy installation** using a compact protocol stack while remaining both **simple** and **flexible**.
- IEEE 802.15.4 is commonly found in the following types of deployments:
 - **Home and building automation.**
 - **Automotive networks.**
 - **Industrial wireless sensor networks.**
 - **Interactive toys and remote controls.**
- Criticisms of IEEE 802.15.4 often focus on its **MAC reliability, unbounded latency, and susceptibility to interference and multi-path fading.**
- **Multi-path fading** refers to multiple copies of the signal hitting the receiver at different points in time because of different signal paths and reflections. The ability to change frequencies can mitigate the effects of multi-path fading.
- The issues of reliability and latency are due to the **Collision Sense Multiple Access/Collision Avoidance (CSMA/CA) algorithm** is usually used. CSMA/CA is an access method in which a device “listens” to make sure no other devices are transmitting before starting its own transmission. If another device is transmitting, a wait time (which is usually random) occurs before “listening” occurs again.
- Interference and multi-path fading occur with IEEE 802.15.4 because it lacks a frequency-hopping technique. This issue is addressed in later variants of 802.15.4 (IEEE 802.15.4e and 802.15.4g).
- IEEE 802.15.4 or IEEE 802.15 Task Group 4 defines **low-data-rate PHY and MAC layer specifications** for wireless personal area networks (WPAN).
- This is a **well-known solution** for low complexity wireless devices with low data rates that need many months or even years of battery life.

- The IEEE 802.15.4 PHY and MAC layers are the foundations for several networking protocol stacks.
- These protocol stacks make use of 802.15.4 at the physical and link layer levels, but the upper layers are different.
- Examples:
 1. **ZigBee:** Promoted through the ZigBee Alliance, ZigBee defines **upper-layer components (network through application) as well as application profiles**. Common profiles include building automation, home automation, and healthcare. ZigBee also defines **device object functions**, such as device role, device discovery, network join, and security.
 2. **6LoWPAN:** 6LoWPAN is an IPv6 adaptation layer defined by the IETF 6LoWPAN working group that describes **how to transport IPv6 packets over IEEE 802.15.4 layers**. RFCs document header compression and IPv6 enhancements to cope with the specific details of IEEE 802.15.4.
 3. **ZigBee IP:** An evolution of the ZigBee protocol stack, ZigBee IP adopts the 6LoWPAN adaptation layer, IPv6 network layer, and RPL routing protocol. In addition, it offers improvements to IP security.
 4. **ISA100.11a:** ISA100.11a is developed by the **International Society of Automation** (ISA) as “Wireless Systems for Industrial Automation: Process Control and Related Applications.” It is based on IEEE 802.15.4-2006, and specifications were published in 2010 and then as IEC 62734. The network and transport layers are based on IETF 6LoWPAN, IPv6, and UDP standards.
 5. **WirelessHART:** WirelessHART, promoted by the HART Communication Foundation, is a protocol stack that offers a **time-synchronized, self-organizing, and self-healing mesh architecture**, leveraging IEEE 802.15.4-2006 over the 2.4 GHz frequency band.
 6. **Thread:** Constructed on top of IETF 6LoWPAN/IPv6, Thread is a protocol stack for a **secure and reliable mesh network** to connect and control products in the home.

1.1 Physical Layer

- The 802.15.4 standard supports an extensive number of PHY options that range from 2.4 GHz to sub-GHz frequencies in ISM bands.
- The original IEEE 802.15.4-2003 standard specified only **three** PHY options based on **direct sequence spread spectrum (DSSS) modulation**.
- DSSS is a modulation technique in which a signal is intentionally spread in the frequency domain, resulting in greater bandwidth.
- The original physical layer transmission options were as follows:

- 2.4 GHz, 16 channels, with a data rate of 250 kbps.
- 915 MHz, 10 channels, with a data rate of 40 kbps.
- 868 MHz, 1 channel, with a data rate of 20 kbps.
- IEEE 802.15.4-2006, 802.15.4-2011, and IEEE 802.15.4-2015 introduced additional PHY communication options, including the following:
 - **OQPSK PHY:** This is DSSS PHY, employing **offset quadrature phase-shift keying** (OQPSK) modulation.
 - * OQPSK is a modulation technique that uses four unique bit values that are signaled by phase changes.
 - * An offset function that is present during phase shifts allows data to be transmitted more reliably.
 - **BPSK PHY:** This is DSSS PHY, employing **binary phase-shift keying** (BPSK) modulation.
 - * BPSK specifies two unique phase shifts as its data encoding scheme.
 - **ASK PHY:** This is **parallel sequence spread spectrum** (PSSS) PHY, employing **amplitude shift keying** (ASK) and BPSK modulation.
 - * PSSS is an advanced encoding scheme that offers increased range, throughput, data rates, and signal integrity compared to DSSS.
 - * ASK uses amplitude shifts instead of phase shifts to signal different bit values.

1.2 Physical Layer Frame

- Fig. 1 shows the frame for the physical layer of IEEE 802.15.4

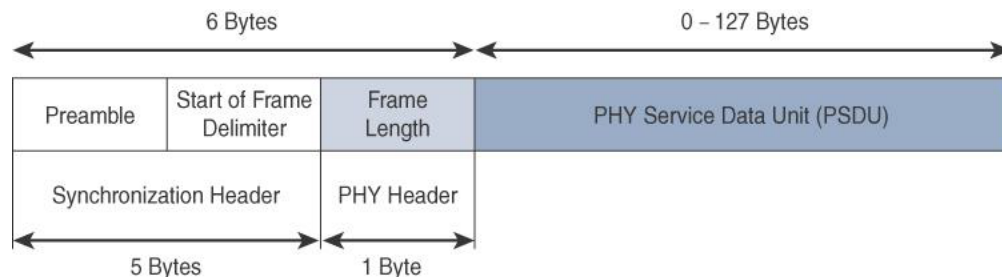


Figure 1: IEEE 802.15.4 PHY Format

- The synchronization header for this frame is composed of the Preamble and the Start of Frame Delimiter fields.
- The Preamble field is a 32 4-byte pattern that identifies the start of the frame and is used to synchronize the data transmission.

- The Start of Frame Delimiter field informs the receiver that the frame contents start immediately after this byte.
- The PHY Header portion of the PHY frame is the frame length value. It lets the receiver know how much total data to expect in the PHY service data unit (PSDU) portion of the 802.4.15 PHY.
- The PSDU is the data field or payload.
- The maximum size of the IEEE 802.15.4 PSDU is 127 bytes. This size is significantly smaller than the lowest MTU setting of other upper-layer protocols, such as IPv6, which has a minimum MTU setting of 1280 bytes. Therefore, fragmentation of the IPv6 packet must occur at the data link layer for larger IPv6 packets to be carried over IEEE 802.15.4 frames.

1.3 MAC Layer

- The IEEE 802.15.4 MAC layer manages access to the PHY channel by defining how devices in the same area will share the frequencies allocated.
- At this layer, the **scheduling, and routing of data frames** are also coordinated.
- The 802.15.4 MAC layer performs the following tasks:
 1. **Network beaconing** for devices acting as coordinators (New devices use beacons to join an 802.15.4 network)
 2. **PAN association and disassociation** by a device
 3. **Device security**
 4. **Reliable link communications** between two peer MAC entities
- The MAC layer achieves these tasks by using various predefined frame types. In fact, **four types of MAC frames** are specified in 802.15.4:
 1. **Data frame:** Handles all transfers of data.
 2. **Beacon frame:** Used in the transmission of beacons from a PAN coordinator.
 3. **Acknowledgement frame:** Confirms the successful reception of a frame.
 4. **MAC command frame:** Responsible for control communication between devices.

1.4 MAC Layer Frame

- MAC frame is carried as the PHY payload.
- Fig. 2 shows the frame format of the four MAC frame types.
- The **MAC Header** field is composed of the Frame Control, Sequence Number, and Addressing fields.

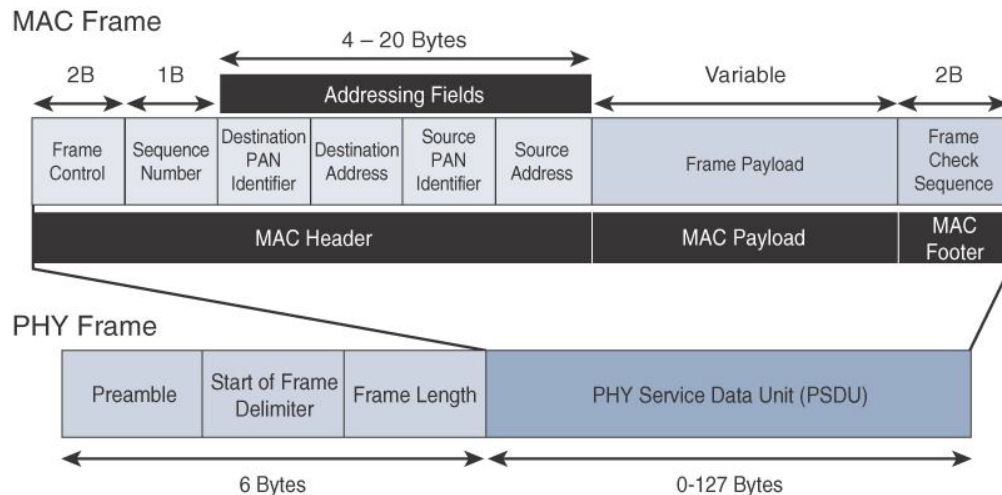


Figure 2: IEEE 802.15.4 MAC Format

- The **Frame Control** field defines attributes such as frame type, addressing modes, and other control flags.
- The **Sequence Number** field indicates the sequence identifier for the frame.
- The **Addressing field** specifies the Source and Destination PAN Identifier fields as well as the Source and Destination Address fields.
- Within the Frame Control portion of the 802.15.4 header is the **Security Enabled field**. When this field is set to a value of 0, the frame format matches Fig. 2. Beginning with the 802.15.4-2006 specification, when this field is set to a value of 1, an Auxiliary Security Header field is added to the 802.15.4 frame, as shown in Fig. 3.

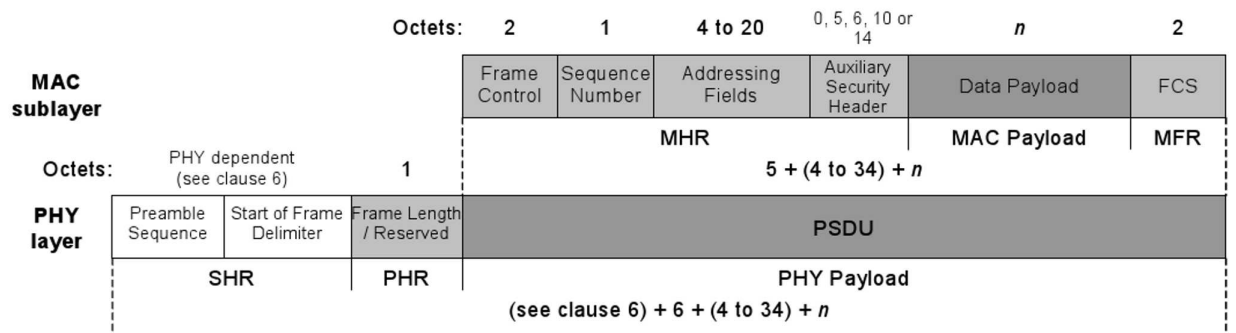


Figure 3: Frame Format with the Auxiliary Security Header Field for 802.15.4-2006 and Later Versions

- The MAC Payload field varies by individual frame type.
- For example, beacon frames have specific fields and payloads related to beacons, while MAC command frames have different fields present.

- The MAC Footer field is nothing more than a frame check sequence (FCS).
- An FCS is a calculation based on the data in the frame that is used by the receiving side to confirm the integrity of the data in the frame.
- IEEE 802.15.4 requires all devices to support a unique 64-bit extended MAC address, based on EUI-64.
- However, because the maximum payload is 127 bytes, 802.15.4 also defines how a 16-bit “short address” is assigned to devices.
- This short address is local to the PAN and substantially reduces the frame overhead compared to a 64-bit extended MAC address.

1.5 Topology

- IEEE 802.15.4-based networks can be built as a star, peer-to-peer, or mesh topology.
- Mesh networks tie together many nodes.
- This allows nodes that would be out of range if trying to communicate directly to leverage intermediary nodes to transfer communications.
- Every 802.15.4 PAN should be set up with a unique ID.
- All the nodes in the same 802.15.4 network should use the same PAN ID.

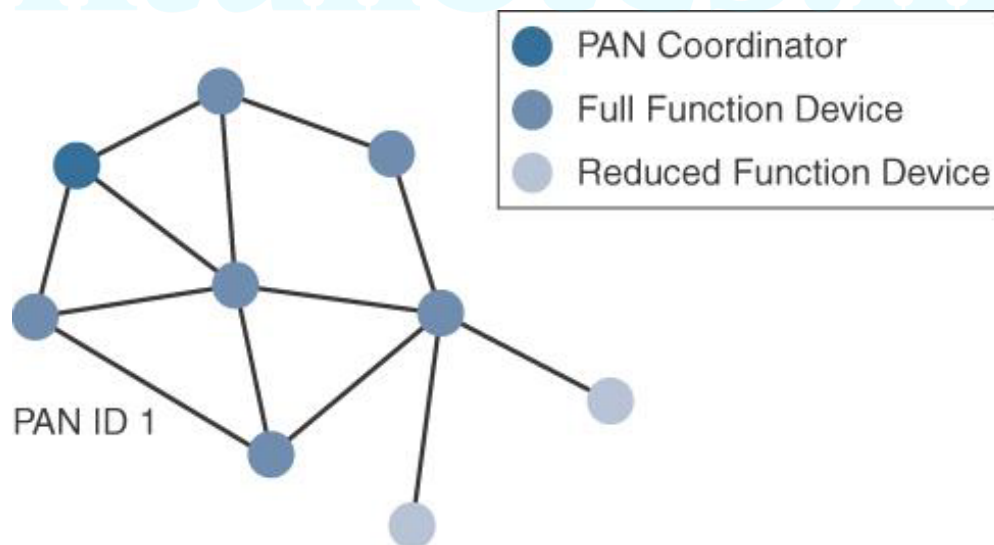


Figure 4: 802.15.4 Sample Mesh Network Topology

- A minimum of one FFD acting as a PAN coordinator is required to deliver services that allow other devices to associate and form a cell or PAN.
- A single PAN coordinator is identified for PAN ID 1.

- FFD devices can communicate with any other devices, whereas RFD devices can communicate only with FFD devices.

1.6 Security

- The IEEE 802.15.4 specification uses **Advanced Encryption Standard** (AES) with a 128-bit key length as the base encryption algorithm for securing its data.
- AES is a **block cipher**, which means it operates on fixed-size blocks of data.
- It is a **symmetric key** means that the same key is used for both the encryption and decryption of the data.
- In addition to encrypting the data, AES in 802.15.4 also validates the data that is sent.
- This is accomplished by a message integrity code (MIC), which is calculated for the entire frame using the same AES key that is used for encryption.
- Using the Security Enabled field in the Frame Control portion of the 802.15.4 header is the first step to enabling AES encryption.
- This field is a single bit that is set to 1 for security.
- Once this bit is set, a field called the **Auxiliary Security Header** is created after the Source Address field, by stealing some bytes from the Payload field.

2 IEEE 802.15.4g and 802.15.4e

- The IEEE 802.15.4e amendment of 802.15.4 2011 expands the MAC layer feature set to remedy the disadvantages associated with 802.15.4, including MAC reliability, unbounded latency, and multi-path fading.
- Also made improvements to better cope with certain application domains, such as **factory and process automation and smart grid**.
- IEEE 802.15.4e 2012 enhanced the IEEE 802.15.4 MAC layer capabilities in the areas of **frame format, security, determinism mechanism, and frequency hopping**.
- IEEE 802.15.4g 2012 is also an amendment to the IEEE 802.15.4 2011 standard.
- The focus of this specification is the smart grid or, more specifically, **smart utility network communication**.
- 802.15.4g seeks to optimize **large outdoor wireless mesh networks for field area networks** (FANs).
- This technology applies to IoT use cases such as the following:

- Distribution automation and industrial supervisory control and data acquisition (SCADA) environments for remote monitoring and control
- Public lighting
- Environmental wireless sensors in smart cities
- Electrical vehicle charging stations
- Smart parking meters
- Microgrids
- Renewable energy

2.1 Physical Layer

- To guarantee interoperability of IEEE 802.15.4g devices, the Wi-SUN Alliance was formed. (SUN stands for smart utility network.)
- In IEEE 802.15.4g-2012, the original IEEE 802.15.4 **maximum PSDU** or payload size of 127 bytes was increased for the SUN PHY to **2047 bytes**.
- This provides a **better match for the greater packet sizes** found in many upper-layer protocols.
- For example, the default IPv6 MTU setting is 1280 bytes. **Fragmentation is no longer necessary** at Layer 2 when IPv6 packets are transmitted over IEEE 802.15.4g MAC frames. Also, the error protection was improved in IEEE 802.15.4g by evolving the CRC from 16 to 32 bits.
- The SUN PHY, as described in IEEE 802.15.4g-2012, supports **multiple data rates** in bands ranging from 169 MHz to 2.4 GHz.
- Within these bands, data must be modulated onto the frequency using at least one of the following PHY mechanisms to be IEEE 802.15.4g compliant:
 - **Multi-Rate and Multi-Regional Frequency Shift Keying (MR-FSK):** Offers good transmit power efficiency due to the constant envelope of the transmit signal.
 - **Multi-Rate and Multi-Regional Orthogonal Frequency Division Multiplexing (MR-OFDM):** Provides higher data rates but may be too complex for low-cost and low-power devices.
 - **Multi-Rate and Multi-Regional Offset Quadrature Phase-Shift Keying (MR-O-QPSK):** Shares the same characteristics of the IEEE 802.15.4-2006 O-QPSK PHY, making multi-mode systems more cost-effective and easier to design.

2.2 MAC layer

The following are some of the main enhancements to the MAC layer proposed by IEEE 802.15.4e-2012:

- **Time-Slotted Channel Hopping (TSCH):**

- TSCH is an IEEE 802.15.4e-2012 MAC operation mode that works to guarantee **media access and channel diversity**.
- Channel hopping, also known as frequency hopping, utilizes **different channels for transmission at different times**.
- TSCH divides time into fixed time periods or “**time slots**” which offer guaranteed bandwidth and predictable latency.
- In a time slot, one packet, and its acknowledgment can be transmitted, **increasing network capacity** because multiple nodes can communicate in the same time slot, using different channels.
- A number of time slots are defined as a “**slot frame**” which is regularly repeated to provide “**guaranteed access**”.
- The transmitter and receiver agree on the channels and the timing for switching between channels through the combination of a global time slot counter and a global channel hopping sequence list, as computed on each node to determine the channel of each time slot.
- TSCH adds robustness in noisy environments and smoother coexistence with other wireless technologies, especially for industrial use cases.

- **Information elements:**

- Information elements (IEs) allow for the exchange of information at the MAC layer in an extensible manner, either as header IEs (standardized) and/or payload IEs (private).
- Specified in a tag, length, value (TLV) format, the IE field allows frames to carry additional metadata to support MAC layer services.
- These services may include IEEE 802.15.9 key management, Wi-SUN 1.0 IEs to broadcast and unicast schedule timing information, and frequency hopping synchronization information for the 6TiSCH architecture.

- **Enhanced beacons (EBs):**

- EBs extend the flexibility of IEEE 802.15.4 beacons to allow the construction of **application-specific beacon content**.
- This is accomplished by **including relevant IEs in EB frames**.
- Some IEs that may be found in EBs include network metrics, frequency hopping broadcast schedule, and PAN information version.

- **Enhanced beacon requests (EBRs):**

- Like enhanced beacons, an enhanced beacon request (EBRs) also leverages IEs.
- The IEs in EBRs allow the sender to **selectively specify the request of information**. Beacon responses are then limited to what was requested in the EBR.
- For example, a device can query for a PAN that is allowing new devices to join or a PAN that supports a certain set of MAC/PHY capabilities.

- **Enhanced Acknowledgement:**

- The Enhanced Acknowledgement frame allows for the integration of a frame counter for the frame being acknowledged.
- This feature helps protect against certain attacks that occur when Acknowledgement frames are spoofed.

- **MAC Frame format:**

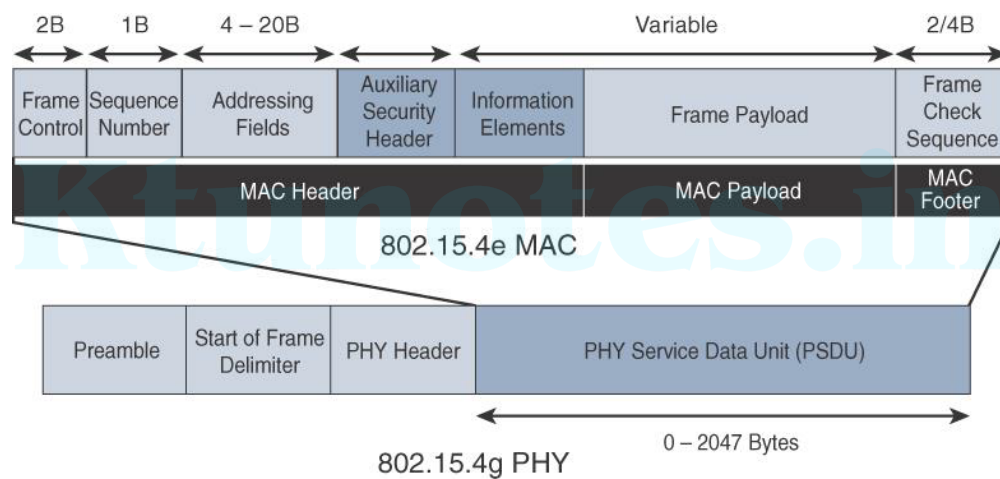


Figure 5: IEEE 802.15.4g/e MAC Frame Format

- The main changes shown in the IEEE 802.15.4e header are the presence of the **Auxiliary Security Header** and **Information Elements** field.
- The Auxiliary Security header provides for the **encryption** of the data frame.
- The IE field contains one or more information elements that allow for **additional information to be exchanged** at the MAC layer.

2.3 Topology

- Deployments of IEEE 802.15.4g 2012 are mostly based on a **mesh topology**.
- Mesh topology is typically the best choice for use cases in the industrial and smart cities areas.

- A mesh topology allows deployments to be done in urban or rural areas, expanding the distance between nodes that can relay the traffic of other nodes.

2.4 Security

- Both IEEE 802.15.4g and 802.15.4e inherit their security attributes from the IEEE 802.15.4 2006 specification.
- Encryption is provided by AES, with a 128-bit key.
- In addition to the **Auxiliary Security Header** field initially defined in 802.15.4 2006, a secure acknowledgment and a **secure Enhanced Beacon** field complete the MAC layer security.

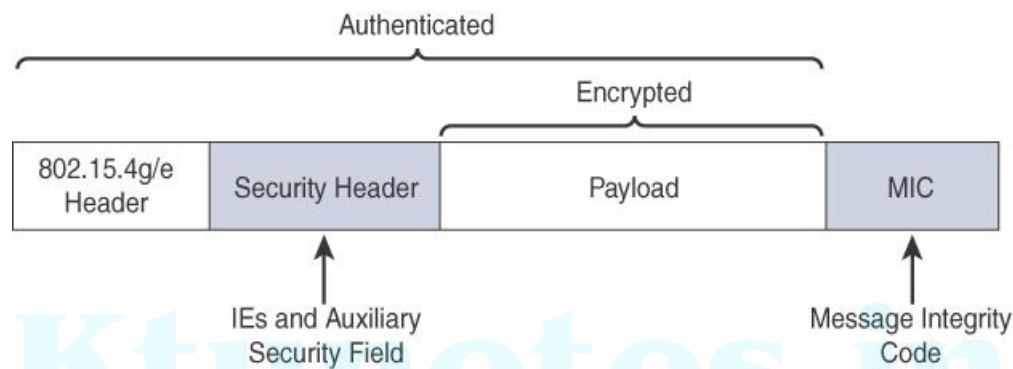


Figure 6: IEEE 802.15.4g/e MAC Layer Security

- The full frame gets authenticated through the **MIC** at the end of the frame.
- The MIC is a **unique value** that is calculated based on the frame contents.
- The Security Header field denoted is composed of the Auxiliary Security field and one or more Information Elements fields.
- Integration of the Information Elements fields allows for the adoption of additional security capabilities, such as the IEEE 802.15.9 Key Management Protocol (KMP) specification.
- KMP provides a means for establishing keys for robust datagram security

3 Modbus

- Modbus is an **industrial protocol** used to monitor and program remote devices via a **master/slave relationship**.
- Modbus is also found in **building management, transportation, and energy applications**.

- Modbus devices communicate using a master-slave technique in which **only one device** (the master) can initiate transactions (called queries).
- The other devices (slaves) respond by supplying the requested data to the master, or by taking the action requested in the query.
- A slave is any peripheral device (I/O transducer, valve, network drive, or another measuring device) that processes information and sends its output to the master using Modbus, while a master can be a host computer running appropriate application software.
- Masters can address individual slaves or can initiate a broadcast message to all slaves. Slaves return a response to all queries addressed to them individually, but do not respond to broadcast queries.

3.1 Transmission Modes

- Standard Modbus networks employ one of **two types of transmission modes**: **ASCII Mode**, or **RTU Mode**.
- The transmission mode defines the bit contents of the message bytes transmitted along the network, and how the message information is to be packed into the message stream and decoded.
- The mode of transmission is usually selected along with other serial port communication parameters (baud rate, parity, etc.) as part of the device configuration. Some devices may only support one mode or the other.
- However, the transmission mode and serial parameters must be the same for all devices connected to a Modbus network.
 - **ASCII Mode (American Standard Code for Information Interchange)**: In the ASCII Transmission Mode (American Standard Code for Information Interchange), each character byte in a message is sent as **2 ASCII characters**.
 - This mode allows time intervals of up to a second between characters during transmission without generating errors.
 - It's most useful when communication is slow.
 - Two ASCII characters are sent as 8-bit data. A start and stop bit are also sent with each message creating a total of **10 bits**. 7 data bits comprise the message and 1 bit is added for either even or odd parity. If no parity is used then the extra stop bit is added to maintain a total transmission of 10 bits. It also uses **LRC (Longitudinal Redundancy Check)** to make sure what we sent is what we received.
 - **RTU mode (Remote Terminal Unit)**: This mode has the advantage of sending **more data in the same amount of time** as in ASCII mode. Each message must be transmitted as a continuous stream of data, however.

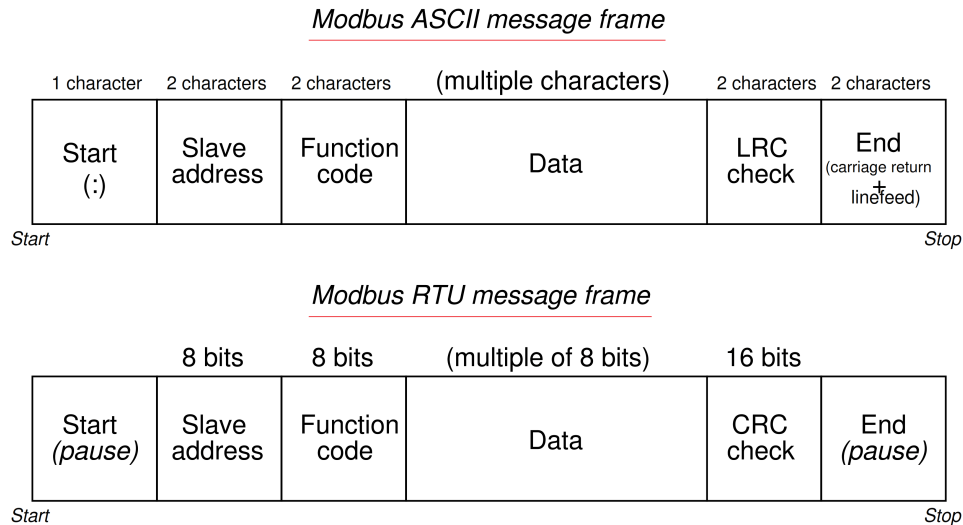


Figure 7: Modbus ASCII and RTU Data Frames

- Each 8-bit message will contain **two 4-bit hex** characters, thus sending the same amount of information in less space. Since we use an extra data-bit (8 vs 7) we send **11 bits** in total. 8 data bits are used and 1 bit is used for either even or odd parity. If no parity is used, then an additional stop bit is added. This brings the total then to 11 bits (1 start + 8 data + 1 parity + 1 stop = 11 bits). This message uses **CRC (Cyclical Redundancy Check)** for error checking.
- ASCII frames require nearly twice (Note 2) the number of bits as RTU frames, making Modbus ASCII slower than Modbus RTU for any given data rate (bits per second).

4 ZigBee

- ZigBee solutions are aimed at **smart objects** and sensors that have **low bandwidth and low power needs**.
- Furthermore, products that are ZigBee compliant and certified by the ZigBee Alliance should interoperate even though different vendors may manufacture them
- The Zigbee specification has undergone several revisions.
- In the 2006 revision, sets of commands and message types were introduced, and increased in number in 2007 (called Zigbee pro) iteration, to achieve different functions for a device, such as metering, temperature, or lighting control.
- The main areas where ZigBee is the most well known include automation for commercial, retail, and home applications and smart energy

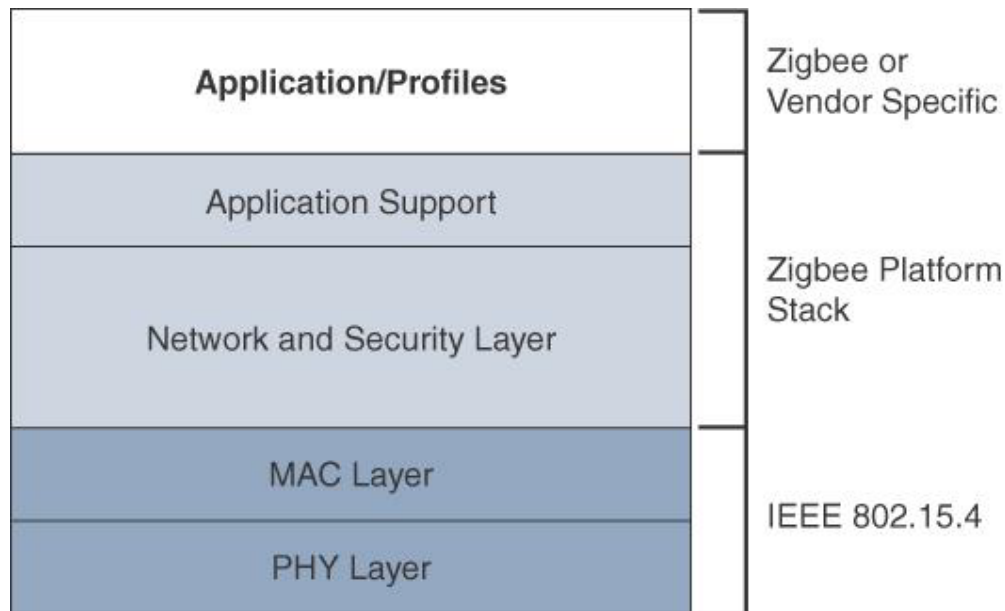


Figure 8: Traditional ZigBee stack

- In the industrial and commercial automation space, ZigBee based devices can handle various functions, from measuring temperature and humidity to tracking assets.
- For home automation, ZigBee can control lighting, ZigBee Smart Energy brings together a variety of interoperable products, such as smart meters, that can monitor and control the use and delivery of utilities, such as electricity and water.
- The ZigBee network and security layer provides mechanisms for network startup, configuration, routing, and secure communications.
- This includes calculating routing paths in what is often a changing topology, discovering neighbors, and managing the routing tables as devices join for the first time.
- The network layer is also responsible for forming the appropriate topology , which is often a mesh but could be a star or tree as well.
- From a security perspective, ZigBee utilizes 802.15.4 for security at the MAC layer, using the Advanced Encryption Standard (AES) with a 128-bit key, and also provides security at the network and application layers.
- ZigBee uses **Ad hoc On-Demand Distance Vector (AODV) routing** across a mesh network.
- Interestingly, this routing algorithm does not send a message until a route is needed.
- Assuming that the next hop for a route is not in its routing table, a network node broadcasts a request for a routing connection.

- This causes a burst of routing-related traffic but after a comparison of various responses, the path with the lowest number of hops is determined for the connection.
- The application support layer interfaces the lower portion of the stack dealing with the networking of ZigBee devices with the higher layer applications.
- ZigBee predefines many application profiles for certain industries, and vendors can optionally create their own custom ones at this layer.
- As mentioned previously, Home Automation and Smart Energy are two examples of popular application profiles.
- ZigBee is one of the most well-known protocols built on an IEEE 802.15.4 foundation. On top of the 802.15.4 PHY and MAC layers, ZigBee specifies its own network and security layer and application profiles. While this structure has provided a fair degree of interoperability for vendors with membership in the ZigBee Alliance, it has not provided interoperability with other IoT solutions.
- To overcome this disadvantage, ZigBee IP was proposed. With ZigBee IP, IEEE 802.15.4 is supported at the physical and MAC layer, and the IP and TCP/UDP protocols and various other open standards are now supported at the network and transport layers. The ZigBee-specific layers are found only at the top of the protocol stack for the applications.

5 LoRaWAN

- LoRaWAN is an **unlicensed-band** LPWA (Low Power Wide Area) technology optimized for **long-range, two-way communications and low power consumption**.
- Published by LoRa Alliance.
- LoRa was a physical layer modulation developed by a French company named Cycleo. Later, Cycleo was acquired by Semtech.
- To differentiate from the physical layer modulation known as LoRa, the LoRa Alliance uses the term LoRaWAN to refer to its architecture and its specifications that describe end-to-end LoRaWAN communications and protocols.
- Figure 9 provides a high-level overview of the LoRaWAN layers. In this figure, it is to be noted that Semtech is responsible for the PHY layer, while the LoRa Alliance handles the MAC layer and regional frequency bands.

5.1 Physical Layer

- Semtech LoRa modulation is based on **chirp spread spectrum modulation**.
- Chirp: Compressed High-Intensity Radar Pulse

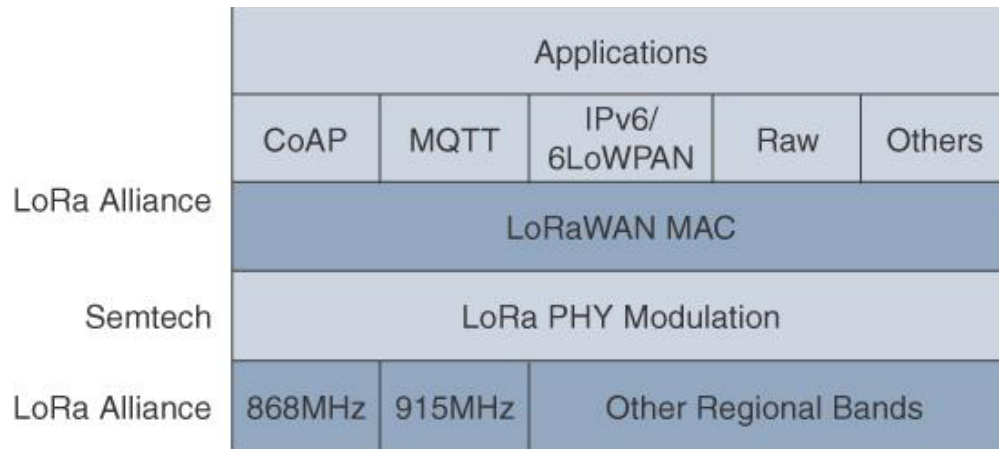


Figure 9: High-level overview of LoRaWAN layers

- **Lower data rate and increase the communication distance.**
- In addition, it allows **demodulation below the noise floor**, offers **robustness to noise and interference**, and manages a **single channel occupation** by different spreading factors. This enables LoRa devices to receive on multiple channels in parallel.
- LoRaWAN uses main unlicensed sub-GHz frequency bands of **433 MHz, 779–787 MHz, 863–870 MHz, and 902–928 MHz**, as well as regional profiles for a subset of the 902–928 MHz bandwidth. For example, Australia utilizes 915–928 MHz frequency bands, while South Korea uses 920–923 MHz and Japan uses 920–928 MHz.
- Understanding LoRa gateways is critical to understanding a LoRaWAN system. A LoRa gateway is deployed as the central hub of a star network architecture. It uses **multiple transceivers and channels** and can demodulate multiple channels at once or even demodulate multiple signals on the same channel simultaneously. LoRa gateways serve as a **transparent bridge relaying data between endpoints**, and the endpoints use a **single-hop wireless connection** to communicate with one or many gateways.
- The data rate in LoRaWAN varies depending on the **frequency bands and adaptive data rate (ADR)**.
- ADR is an algorithm that manages the data rate and radio signal for each endpoint.
- The ADR algorithm ensures that packets are delivered at the best data rate possible and that network performance is both optimal and scalable.
- Endpoints close to the gateways with good signal values transmit with the highest data rate, which enables a shorter transmission time over the wireless network, and the lowest transmit power.
- Meanwhile, endpoints at the edge of the link budget communicate at the lowest data rate and highest transmit power.

- Usually, ADR is used for fixed endpoints and a fixed data rate is used for mobile endpoints.
- An important feature of LoRa is its **ability to handle various data rates** via the spreading factor. Devices with a low spreading factor (SF) achieve less distance in their communications but transmit at faster speeds, resulting in less airtime. A higher SF provides slower transmission rates but achieves higher reliability at longer distances.

5.2 MAC Layer

- The MAC layer is defined in the LoRaWAN specification.
- This layer takes advantage of the LoRa physical layer and classifies LoRaWAN endpoints to **optimize their battery life** and ensure downstream communications to the LoRaWAN endpoints.
- The LoRaWAN specification documents three classes of LoRaWAN devices:
 - **Class A:** This class is the **default** implementation. Optimized for **battery-powered** nodes, it allows **bidirectional communications**, where a given node is able to receive downstream traffic after transmitting. Two receive windows are available after each transmission.
 - **Class B:** This class was designated “experimental” in LoRaWAN 1.0.1 until it can be better defined. A Class B node or endpoint should get **additional receive windows** compared to Class A, but gateways must be synchronized through a beaconing process.
 - **Class C:** This class is particularly adapted for **powered nodes**. This classification enables a node to be **continuously listening** by keeping its receive window open when not transmitting.
- Frame format: LoRaWAN messages, either uplink or downlink, have a PHY payload composed of a **1-byte MAC header**, a **variable byte MAC payload**, and a **MIC that is 4 bytes** in length.

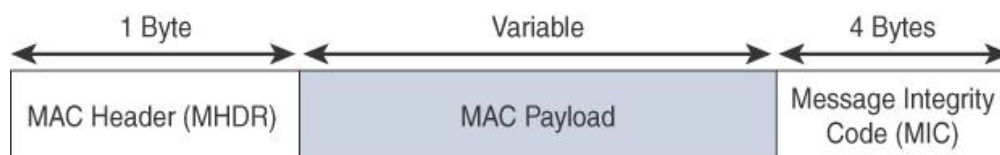


Figure 10: High-Level LoRaWAN MAC Frame Format

- The MAC payload size depends on the **frequency band and the data rate**, ranging from 59 to 230 bytes for the 863-870 MHz band and 19 to 250 bytes for the 902-928 MHz band.
- In version 1.0.x, LoRaWAN utilizes six MAC message types.

- LoRaWAN devices use **join request** and **join accept** messages for activation and joining the network.
- The other message types are **unconfirmed data up/down** and **confirmed data up/down**.
- A “confirmed” message is one that must be acknowledged, and “unconfirmed” signifies that the end device does not need to acknowledge.
- “up/down” is simply a directional notation identifying whether the message flows in the uplink or downlink path.
- Uplink messages are sent from endpoints to the network server and are relayed by one or more LoRaWAN gateways.
- Downlink messages flow from the network server to a single endpoint and are relayed by only a single gateway.
- LoRaWAN endpoints are uniquely addressable through a variety of methods, including the following:
 - An endpoint can have a **global end device ID** or DevEUI represented as an IEEE EUI-64 address.
 - An endpoint can have a **global application ID** or AppEUI represented as an IEEE EUI-64 address that uniquely identifies the application provider, such as the owner, of the end device.
- In a LoRaWAN network, endpoints are also known by their **end device address**, known as a DevAddr, a 32-bit address.
- The 7 most significant bits are the network identifier (NwkID), which identifies the LoRaWAN network.
- The 25 least significant bits are used as the network address (NwkAddr) to identify the endpoint in the network.

5.3 Topology

- LoRaWAN topology is often described as a “**star of stars**” topology. The infrastructure consists of endpoints exchanging packets through gateways acting as bridges, with a central LoRaWAN network server. Gateways connect to the backend network using standard IP connections, and endpoints communicate directly with one or more gateways.
- LoRaWAN endpoints **transport their selected application data** over the LoRaWAN MAC layer on top of one of the supported PHY layer frequency bands. The application data is contained in upper protocol layers.

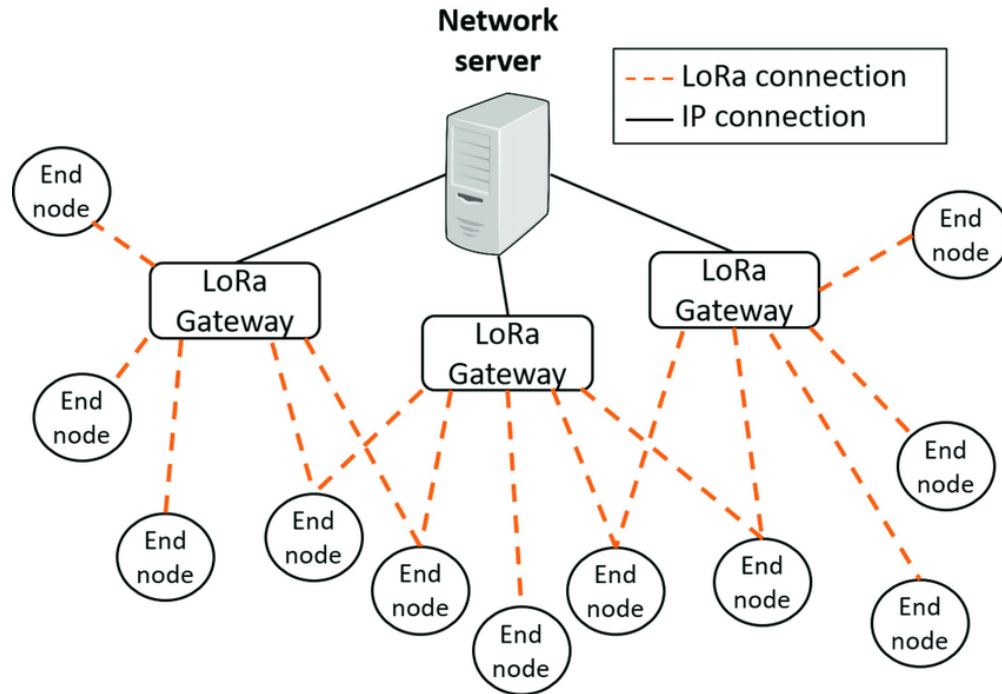


Figure 11: LoRaWAN Topology

- LoRaWAN gateways act as **bridges that relay between endpoints and the network servers**. Multiple gateways can receive and transport the same packets. When duplicate packets are received, the network server performs **de-duplication**.
- The LoRaWAN network server **manages the data rate and radio frequency (RF)** of each endpoint through the adaptive data rate (ADR) algorithm. ADR is a key component of the endpoints' network scalability, performance, and battery life. The LoRaWAN network server **forwards application data** to the application servers.

5.4 Security

- Security in a LoRaWAN deployment applies to different components of the architecture.
- LoRaWAN endpoints must implement **two layers** of security, protecting communications and data privacy across the network.
- The first layer called **“network security”** but applied at the MAC layer, guarantees the authentication of the endpoints by the LoRaWAN network server.
- Also, it protects LoRaWAN packets by performing encryption based on AES.
- Each endpoint implements a **network session key** (NwkSKey), used by both itself and the LoRaWAN network server.

- The NwkSKey ensures data integrity by computing and checking the MIC of every data message, and encrypting and decrypting MAC-only data message payloads.
- The second layer is an **application session key** (AppSKey), which performs encryption and decryption functions between the endpoint and its application server.
- Furthermore, it computes and checks the application-level MIC, if included. This ensures that the LoRaWAN service provider does not have access to the application payload if it is not allowed that access.

6 LTE-M (LTE-MTC)

- LTE-M stands for Long Term Evolution - Machine Type Communication.
- LTE-M is a **low power wide area** (LPWA) technology standard published by **3GPP**.
- **3rd Generation Partnership Project** (3GPP) is a standards organization that unites multiple telecommunications standards development organizations to provide a stable environment to produce the reports and specifications that define 3GPP technologies.
- Existing cellular technologies, such as GPRS, Edge, 3G, and 4G/LTE, are not particularly well adapted to battery-powered devices and small objects specifically developed for the Internet of Things.
- This has resulted in LTE-M that uses **cellular technologies** to address specific IoT requirements, such as **low throughput** and **low power consumption** with decreased complexity and cost of the LTE devices.
- **Features of LTE-M:**
 1. **Lower receiver bandwidth:** Bandwidth has been lowered to 1.4 MHz versus the usual 20 MHz. This further simplifies the LTE endpoint.
 2. **Lower data rate:** Data is around 200 kbps for LTE-M, compared to 1 Mbps for Cat 0.
 3. **Half-duplex mode:** LTE-M offers a half-duplex mode that decreases node complexity and cost.
 4. **Enhanced discontinuous reception (eDRX):** This capability increases from seconds to minutes the amount of time an endpoint can “sleep” between paging cycles. A paging cycle is a periodic check-in with the network. This extended “sleep” time between paging cycles extends the battery life for an endpoint significantly.
 5. **Wide range of devices:** An LTE-M-based system is able to support a wide variety of cellular devices designed for many different purposes; ranging from smart meters to vending machines and automotive fleet management to safety

and medical devices. As these devices differ from use to use, they have many disparate requirements, therefore any LTE machine-to-machine deployment must be flexible.

6. **Low device cost:** Most IoT-enabled devices must be small and fit on equipment that is very cost-sensitive. With many low-cost IoT systems already available, LTE-M provides the benefits of a cellular system, but at a significantly lower cost.
 7. **Long battery life:** IoT deployment devices will need to be left unattended for long periods of time in areas where there is no power supply. Constantly replacing batteries is expensive, so any device should have a battery life of up to ten years. This means that the system must be very effective in terms of battery economy. LTE-M allows a battery life of as long as 10 years or more for a wide range of use cases, with the modem costs reduced to 20-25% of the current modems.
 8. **Improved coverage:** Applications for this network should operate within a variety of locations, not just where coverage is good. They will need to operate within buildings, often in locations with restrained access or even underground (think smart meters). With its deep signal penetration capabilities, LTE-M provides a perfect answer to all these needs.
- **Applications:** LTE-M enables a wide variety of applications that require low data rates and real-time communications. Some of the potential use cases of LTE-M are:
 1. **Smart cities:** intelligent public lighting, monitoring environmental conditions, checking which parking lots are free, etc.
 2. **Smart meters:** measuring autonomously the consumption of energy, water, or natural gas using IoT protocols and even in hard-to-reach places where signal strength is next to nothing.
 3. **Automotive and logistics:** Fleet control, route optimization to determine the exact location of the driver or to minimize incidences of losses of goods, etc.
 4. **Connected Medicine:** Connected sanitary and medical devices.
 5. **Personal IoT devices:** People and animal trackers, fitness wearables, smart-watches, etc.

7 NB-IoT

- NB-IoT stands for narrowband IoT.
- It is a 3GPP radio technology standard that addresses the requirements of the IoT.
- It provides improved indoor coverage, support of massive number of low throughput devices, low delay sensitivity, low device cost, low device power consumption, and optimized network structure.

- **Operation Modes:** NB-IoT technology occupies a frequency band of 180 kHz bandwidth, which corresponds to one resource block in LTE transmission. With this selection, the following operation modes are possible:
 1. **Standalone:** A GSM carrier is used as an NB-IoT carrier, enabling reuse of 900 MHz or 1800 MHz.
 2. **In-band:** Part of an LTE carrier frequency band is allocated for use as an NB-IoT frequency. The service provider typically makes this allocation, and IoT devices are configured accordingly.
 3. **Guard band:** An NB-IoT carrier is between the LTE or WCDMA bands. This requires coexistence between LTE and NB-IoT bands.

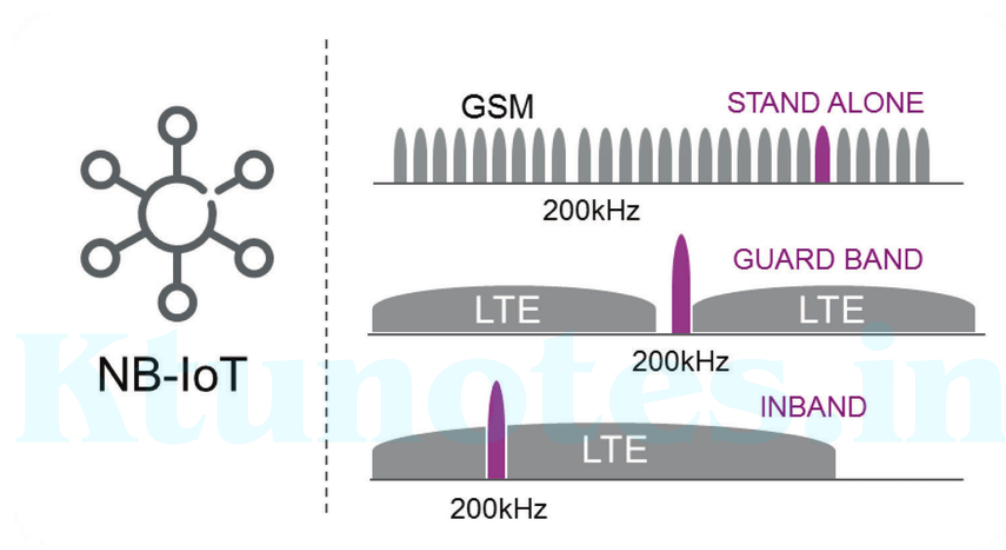


Figure 12: NB-IoT Operation Modes

8 IP as the IoT Network Layer

- IP has become the de-facto standard in many areas of IoT.
- **Need for IP optimization:** The IoT is largely built on the Internet Protocol suite. In addition to coping with the integration of non-IP devices, we may need to deal with the limits at the device and network levels that IoT often imposes. Therefore, optimizations are needed at various layers of the IP stack to handle the restrictions that are present in IoT networks. There are three main areas to focus on for optimization. They are:
 1. **Constrained nodes:** Nodes with sensors/actuators designed to handle a specific application. The main challenges that push to optimize IP for constrained nodes are:

- **Different characteristics:** Depending on its functions in an IoT network, a “thing” may or may not offer similar characteristics compared to a generic PC or server.
 - **Limited throughput:** The network protocol stack on an IoT node may be required to communicate through an **unreliable path**. Even if a full IP stack is available on the node, this causes problems such as **limited or unpredictable throughput and low convergence** when a topology change occurs.
 - **Power consumption:** Many IoT devices are battery-powered, and the battery life varies from a few months to 10+ years. These power consumption requirements on battery-powered nodes impact **communication intervals** also.
2. **Constrained Networks:** Low power and lossy networks. *Lossy* in this context refers to network unreliability that is caused by disruptions in the data flow or packet loss. The constrained networks are limited by the following challenges that make IP optimization necessary for constrained networks.
- **Low power**
 - **Limited bandwidth**
 - **Fluctuations in the packet delivery rate:** The packet delivery rate (PDR) may oscillate between low and high percentages. Large bursts of unpredictable errors and even loss of connectivity at times may occur.
 - **Latency and control plane reactivity:** One of the golden rules in a constrained network is to “underreact to failure.” Due to the low bandwidth, a constrained network that overreacts can lead to a network collapse.
 - **Control plane traffic:** Control plane traffic must also be kept at a minimum; otherwise, it consumes the bandwidth that is needed by the data traffic.
3. **IP versions:** There are two IP versions - IPv4 and IPv6. Since IoT is a collection of different devices that use different protocols and different standards, it should support both versions concurrently. To ensure interoperability, techniques like **Tunneling** and **Translation** need to be employed. (Note: Tunneling is a protocol that allows for the secure movement of data from one network to another. Translation is a protocol for enhanced user experience.)

8.1 Optimizing IP for IoT

- The use of constrained nodes and constrained networks mandate optimization at various layers and on multiple protocols of the IP architecture.
- Normally the IP is optimized for IoT by using an **adaptation layer**.
- The adaptation layer **removes the compatibility issues** between IPv6 and IEEE802.15.4 for efficient transmission of packets.
- It also **reduces the transmission overhead** of packets.

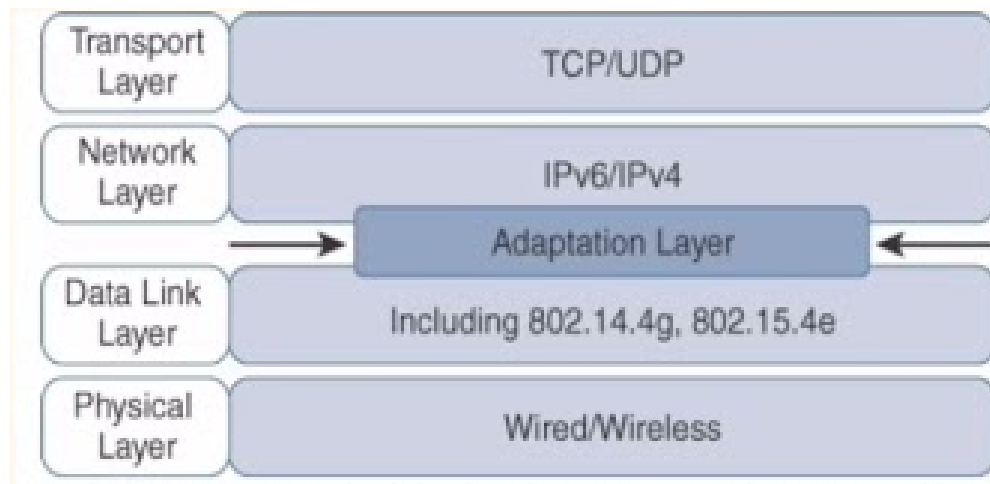


Figure 13: Optimizing IP for IoT using an adaptation layer

- **Physical layer** provides the basic communication in the network. It **transmits the data bits** over 802.15.4 by converting them into signals.
- The task of the **Data link layer** is to detect and correct the errors which may occur during the transmission of data bits. The Medium access layer or MAC layer is present in the data link layer. This layer senses the medium for the **collision-free transmission of frames** using various protocols like CSMA/CD or CSMA/CA.
- **Adaptation layer** is present between the network layer and the data link layer. Large-sized headers like IPv6, UDP, and ICMP are **compressed** by this layer. Also, it handles the **fragmentation and reassembly of packets**.
- Above the adaptation layer there is a **network layer** which provides **routing of the packets**. It uses networking protocol i.e. IP (Internet Protocol) for providing addresses to the nodes.
- **Communication between nodes** could be connectionless or connection-oriented for which the **transport layer** is responsible.
- **Functions of the adaptation layer:** 6LoWPAN expresses each capability in a self-contained sub-header: mesh addressing, fragmentation, and header compression.

1. Header Compression:

- IPv6 packet size is 1280 bytes. However, the maximum MAC frame size defined by IEEE 802.15.4 is 127 bytes.
- Of these 127 bytes, 21 bytes are for the MAC frame header, 1 byte is for the 6LoWPAN header, 40 bytes are for the IPv6 header, 8 bytes are for the UDP header and 4 bytes are for the Frame Check Sequence. i.e., only 53 bytes are left for the actual payload. The rest of the space is used by the headers.

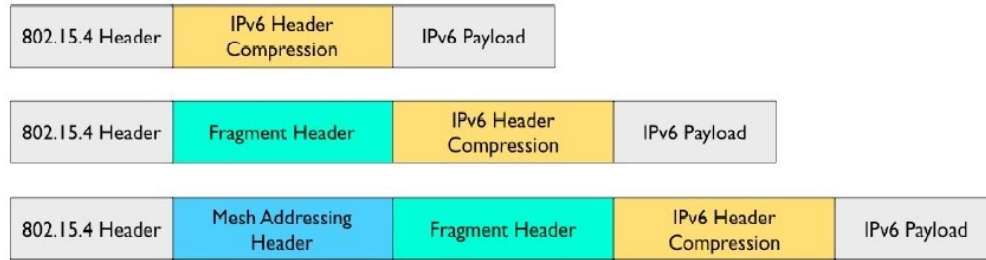


Figure 14: 6LoWPAN header stacks

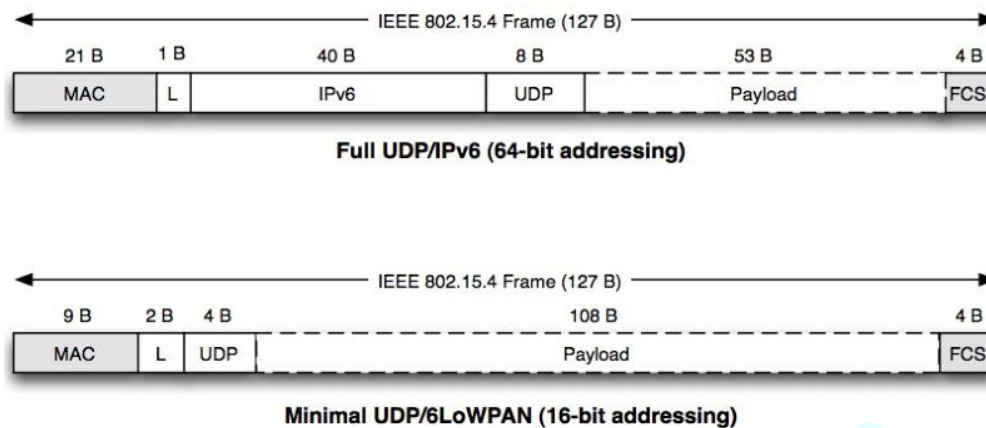


Figure 15: 6LoWPAN Header Compression

- Hence, it is necessary to provide efficient header compression (of IPv6 and UDP), without losing information in order to adapt 1280 bytes of IPv6 MTU into 127 bytes of 802.15.4 frames.
- 6LoWPAN provides a **stateless compression** of the IPv6 header: all the information to compress and decompress is contained in the packet. The sender/receiver does not keep any information.
- It is a **flow-independent compression**, each packet is compressed independently from the others.
- Various techniques have been suggested to perform compression of the header so as to create effective space for the payload in 6LoWPAN frame. **HC1** (Header Compression 1) and **IPHC** (Internet Protocol Header Compression) are two well-known techniques among them.
- Both HC1 and IPHC compress 40 bytes of IPv6 header to 2 bytes which indicates the way the IPv6 header is compressed and from where its values can be recovered.
- In the process of compression, the header fields whose values could be derived again at some other layer is elided from the main header.
- When the compressed header along with the frame reaches the destination, the elided header fields are derived by the adaptation layer. These fields are mainly derived from the link layer header which is carried in an 802.15.4

frame.

- At the top of Fig. 15, a 6LoWPAN frame without any header compression is presented: The full 40-byte IPv6 header and the 8-byte UDP header are visible. The 6LoWPAN header is only a single byte in this case. It is clear that the uncompressed IPv6 and UDP headers leave only 53 bytes of data payload out of the 127-byte maximum frame size in the case of IEEE 802.15.4.
- The bottom half of Fig. 15 shows a frame where header compression has been enabled for a best-case scenario. The 6LoWPAN header increases to 2 bytes to accommodate the compressed IPv6 header, and UDP has been reduced in half, to 4 bytes from 8. Most importantly, the header compression has allowed the payload to more than double, from 53 bytes to 108 bytes, which is obviously much more efficient.

2. Fragmentation and Reassembly of packets:

- Maximum transmission unit (MTU) of IPv6 packets is 1280 bytes whereas in 802.15.4 the permissible packet size is only 127 bytes.
- Hence to encapsulate the IPv6 large-sized packets onto 802.15.4, they need to be fragmented, transmitted, and reassembled after reaching the destination.
- At the sender node when an IPv6 packet size exceeds the available link-layer payload size, the 6LoWPAN adaptation layer iteratively breaks the IPv6 packets into fragments. The size of the fragments will be according to the maximum frame size at the data link layer. Each fragment will then include a fragment header before it is transmitted.

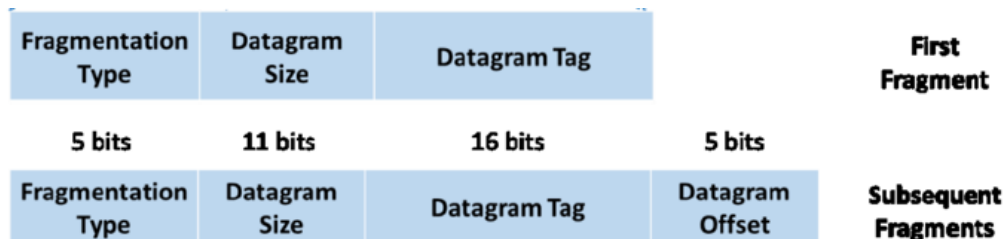


Figure 16: 6LoWPAN Fragmentation Header

- As indicated by Fig. 16The fragment header utilized by 6LoWPAN is composed of three primary fields: **Datagram Size**, **Datagram Tag**, and **Datagram Offset**. The Datagram Size field specifies the total size of the unfragmented payload. The Datagram Tag identifies the set of fragments for a payload. Finally, the Datagram Offset field delineates how far into a payload a particular fragment occurs.
- In Fig. 16, the 6LoWPAN fragmentation header field itself uses a unique bit value to identify that the subsequent fields behind it are fragment fields as opposed to another capability, such as header compression.
- Also, in the first fragment, the Datagram Offset field is not present because it would simply be set to 0.

3. Mesh Addressing:

- The purpose of the 6LoWPAN mesh addressing function is to **forward packets over multiple hops**.
- Three fields are defined for this header: **Hop Limit, Source Address, and Destination Address**.
- The hop limit provides an upper limit on how many times the frame can be forwarded. Each hop decrements this value by 1 as it is forwarded. Once the value hits 0, it is dropped and no longer forwarded.
- The Source Address and Destination Address fields are IEEE 802.15.4 addresses indicating the endpoints of an IP hop.

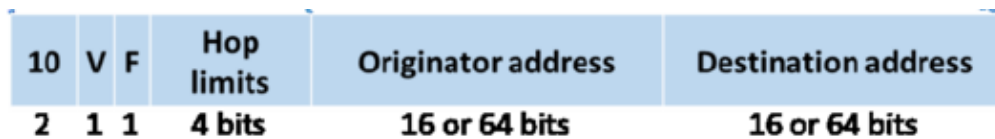


Figure 17: 6LoWPAN Mesh Addressing Header

- In 6LoWPAN, the adaptation layer is also responsible for routing the fragmented packets (mesh-under routing).
- If a packet is fragmented by the adaptation layer then the mesh addressing header is sent along with every fragment. The router may route the fragments of the same packet through different routes. When all fragments reach the destination node, then they are reassembled by the adaptation layer present there. If any of the fragments belonging to a packet is missing, then all fragments are to be retransmitted.

8.2 RPL

- Low-power and Lossy Networks (LLNs) consist largely of constrained nodes (with **limited processing power, memory, and sometimes energy** when they are battery-operated or energy scavenging). These nodes are interconnected by lossy links, typically supporting only **low data rates**, that are usually **unstable** with relatively **low packet delivery rates**.
- Another characteristic of such networks is that the traffic patterns are not simply point-to-point (between devices inside the LLN) but in many cases point-to-multipoint (from a central control point to a subset of devices inside the LLN) or multipoint-to-point (from devices inside the LLN towards a central control point). Such networks may potentially comprise up to thousands of nodes.
- **IPv6 Routing Protocol for Low Power and Lossy Networks (RPL)** is considered the defacto routing protocol for IoT.
- It provides a mechanism whereby multipoint-to-point traffic from devices inside the LLN towards a central control point, as well as point-to-multipoint traffic from the

central control point to the devices inside the LLN, is supported. Support for point-to-point traffic is also available.

- RPL is a **distance vector routing protocol**, here the routing is based on Destination Oriented Directed Acyclic Graphs (DODAGs).
- In an RPL network, each node acts as a router and becomes part of a mesh network.
- Routing is performed at the IP layer. Each node examines every received IPv6 packet and determines the next-hop destination based on the information contained in the IPv6 header. No information from the MAC-layer header is needed to perform the next-hop determination. This is referred to as mesh over routing.
- To cope with the constraints of computing and memory that are common characteristics of constrained nodes, the protocol defines two modes:
 - **Storing mode:** All nodes contain the full routing table of the RPL domain. Every node knows how to directly reach every other node.
 - **Non-storing mode:** Only the border router(s) of the RPL domain contain(s) the full routing table. All other nodes in the domain only maintain their list of parents and use this as a list of default routes toward the border router. This abbreviated routing table **saves memory space and CPU**. When communicating in non-storing mode, a node always forwards its packets to the border router, which knows how to ultimately reach the final destination.
 - All nodes except the root can be either in storing or non-storing mode. The root is always in storing mode.
- **Basic Terminology:**
 1. **Directed Acyclic Graph (DAG):** DAG is a directed graph where **no cycles** exist. This means that from any vertex or point in the graph, one cannot follow an edge or a line back to this same point. All of the edges are arranged in paths oriented toward and terminating at one or more root nodes. Fig. 18 shows a basic DAG.

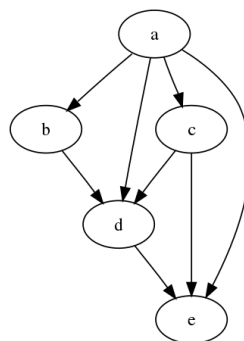


Figure 18: Example of a Directed Acyclic Graph (DAG)

2. **Root:** A DAG root is a node within the DAG that has no outgoing edge. Because the graph is acyclic, by definition all DAGs must have at least one DAG root and all paths terminate at a DAG root. The DODAG root may act as a border router for the DODAG; in particular, it may aggregate routes in the DODAG and may redistribute DODAG routes into other routing protocols.
 3. **Up:** It is any edge that is directed towards the root.
 4. **Down:** It is any edge that is directed away from the root.
 5. **Destination Oriented DAG (DODAG):** A DAG rooted at a single destination, i.e. at a single DAG root (the DODAG root) with no outgoing edges.
 6. **Rank:** A node's Rank defines the node's individual position relative to other nodes with respect to a DODAG root. In other words, it is the distance from the root. Rank strictly increases in the Down direction and strictly decreases in the Up direction.
 7. **Objective Function (OF):** An objective function (OF) defines how metrics are used to select routes and establish a node's rank.
 8. **RPL Instance:** A set of one or more DODAGs that share an objective function. An RPL node can belong to at most one DODAG in an RPL Instance. Each RPL Instance operates independently of other RPL Instances.
 9. **DODAG ID** Each DODAG has a unique IPv6 ID (128 bits) that is given to only the root. In other words, it is the identifier of a DODAG root. The tuple (RPLInstanceID, DODAGID) uniquely identifies a DODAG.
 10. **DODAG parent:** A parent of a node within a DODAG is one of the immediate successors of the node on a path towards the DODAG root. A DODAG parent's Rank is lower than the node's.
 11. **Host, Router and Node:** As they form networks, LLN devices often mix the roles of host and router when compared to traditional IP networks. A "host" refers to an LLN device that can generate but does not forward RPL traffic and a "router" refers to an LLN device that can forward as well as generate RPL traffic. A "node" refers to any RPL device, either a host or a router.
 12. **Goal:** The Goal is an application-specific goal that is defined outside the scope of RPL. Any node that roots a DODAG will need to know about this Goal to decide if the Goal can be satisfied or not. A typical Goal is to construct the DODAG according to a specific objective function and to keep connectivity to a set of hosts (e.g. to use an objective function that minimizes a metric and to be connected to a specific database host to store the collected data).
 13. **Grounded:** A DODAG is grounded when the DODAG root can satisfy the Goal.
 14. **Floating:** A DODAG is floating if it is not Grounded. A floating DODAG is not expected to have the properties required to satisfy the goal. It may, however, provide connectivity to other nodes within the DODAG.
- **Overview of the protocol:**

- In a DODAG, each node maintains up to three parents that provide a path to the root. Typically, one of these parents is the **preferred parent**, which means it is the preferred next hop for upward routes toward the root.
- The **routing graph** created by the set of DODAG parents across all nodes defines the full set of upward routes. RPL protocol implementation should ensure that routes are loop-free by disallowing nodes from selected DODAG parents that are positioned further away from the border router.
- Upward routes in RPL are discovered and configured using **DAG Information Object (DIO)** messages. Nodes listen to DIOs to handle changes in the topology that can affect routing. The information in DIO messages determines **parents and the best path to the DODAG root**.
- When no DIO is received, and if a node wants to join a DODAG, it sends the **DODAG Information Solicitation (DIS)** message to see if any DODAG exists.
- Nodes establish downward routes by advertising their parent set toward the DODAG root using a **Destination Advertisement Object (DAO)** message. DAO messages allow nodes to inform their parents of their presence and reachability to descendants. In response to the DAO, the root or parent sends a **DAO Acknowledgement (DAO-Ack)** to the child.

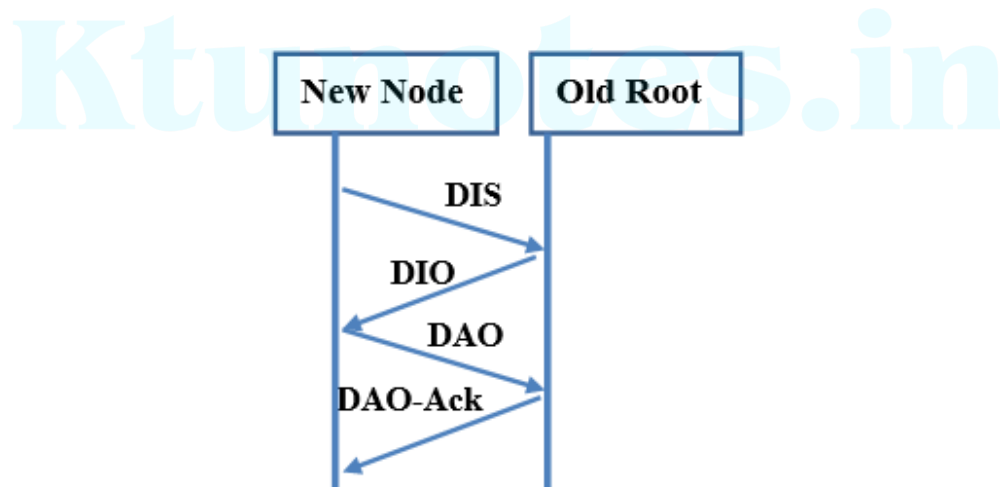


Figure 19: RPL Control Messages

- In the case of the non-storing mode of RPL, nodes sending DAO messages report their parent sets directly to the DODAG root (border router), and only the root stores the routing information. The root uses the information to then determine source routes needed for delivering IPv6 datagrams to individual nodes downstream in the mesh.
- For storing mode, each node keeps track of the routing information that is advertised in the DAO messages. While this is more power- and CPU-intensive for each node, the benefit is that packets can take shorter paths between destinations in

the mesh. The nodes can make their own routing decisions; in non-storing mode, on the other hand, all packets must go up to the root to get a route for moving downstream.

- RPL messages, such as DIO and DAO, run on top of IPv6. These messages exchange and advertise downstream and upstream routing information between a border router and the nodes under it.
- **DODAG Formation:** Root in a DODAG is a special node. All nodes don't have the capability to be roots. A root is programmed that way. Suppose there are 5 nodes : A, B, C, D, and E. Then, to make a DODAG, the following steps are taken:
 1. A multicasts DIOs.
 2. Rest of the nodes (B,C,D and E) upon receiving DIOs will try to join regardless of their distances. Upon receiving the DIOs they also come to know that their distances from the A are 1,1,3,4 Respectively.
 3. Then B, C, D, and E send DAOs to A.
 4. A accepts them by replying with DAO-Ack. These steps are shown in Fig. 20.

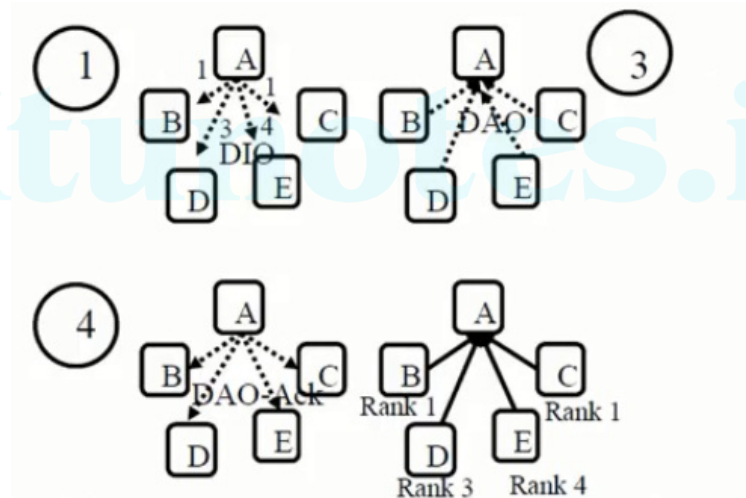


Figure 20: DODAG Formation

5. Now another round begins, and it starts with the nodes that are nearest to the root. Since B & C have a rank of 1, they start sending DIOs.
6. D receives these DIOs and figures out that its distances from B & C are 1 & 2 respectively.
7. Similarly E receives these DIOs , and it figures out its distances from B & C to be equal to 2 & 1 respectively .
8. Since D is closer to B and E is closer to C, D Sends DAO to B and E sends DAO to C.

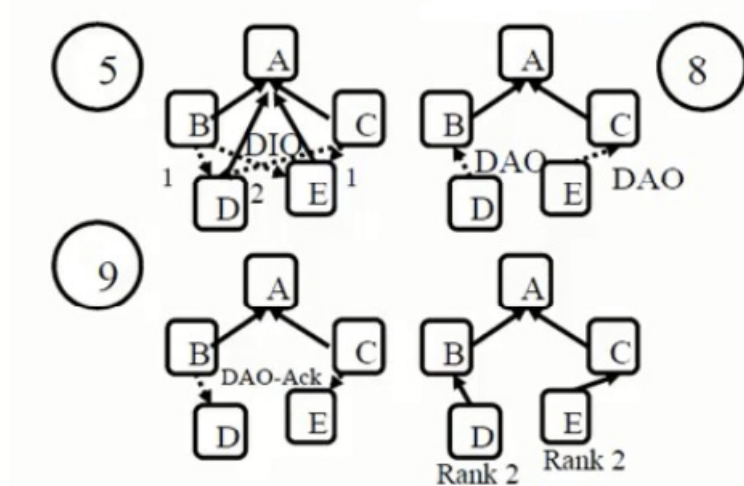


Figure 21: DODAG Formation (contd)

9. To finalize the DODAG formation, B sends a DAO-Ack to D, and C sends a DAO-Ack to E. These steps are shown in Fig. 21.
- **Metrics:** RPL defines a large and flexible set of metrics and constraints for routing. Some of the RPL routing metrics and constraints include the following:
 1. **Expected Transmission Count (ETX):** Assigns a discrete value to the number of transmissions a node expects to make to deliver a packet.
 2. **Hop Count:** Tracks the number of nodes traversed in a path. Typically, a path with a lower hop count is chosen over a path with a higher hop count.
 3. **Latency:** Varies depending on power conservation. Paths with a lower latency are preferred.
 4. **Link Quality Level:** Measures the reliability of a link by taking into account packet error rates caused by factors such as signal attenuation and interference.
 5. **Link Color:** Allows manual influence of routing by administratively setting values to make a link more or less desirable. These values can be either statically or dynamically adjusted for specific traffic types.
 6. **Node State and Attribute:** Identify nodes that function as traffic aggregators and nodes that are being impacted by high workloads. High workloads could be indicative of nodes that have incurred high CPU or low memory states. Naturally, nodes that are aggregators are preferred over nodes experiencing high workloads.
 7. **Node Energy:** Avoids nodes with low power, so a battery-powered node that is running out of energy can be avoided and the life of that node and the network can be prolonged.
 8. **Throughput:** Provides the amount of throughput for a node link. Often, nodes conserving power use lower throughput. This metric allows the prioritization of paths with higher throughput.